



INSTITUTE OF SPACE TECHNOLOGY
KICSIT, Kahuta Campus



Course Title: Adv Database Lab

Project Proposal

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BSCS-4B

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PROJECT PROPOSAL

Advanced Hospital Management System Using Flask and MongoDB

1. Introduction

Healthcare institutions require an efficient system to manage patient and doctor records accurately. Traditional paper-based record management systems are time-consuming, difficult to maintain, and prone to human errors. Hospitals often face challenges such as data duplication, loss of records, and inefficient patient management.

The Advanced Hospital Management System is a web-based application developed using Flask and MongoDB to automate and simplify hospital management processes. The system allows administrators to manage doctor information, patient records, doctor assignments, and search operations through a centralized dashboard. The project aims to provide a secure, organized, and efficient environment for handling hospital data digitally.

2. Problem Statement

Many hospitals and clinics still rely on manual record-keeping methods. These methods create several problems, including:

- Difficulty in maintaining patient records.
- Slow retrieval of information.
- Increased chances of data loss.
- Record duplication and inconsistency.
- Time-consuming administrative tasks.
- Difficulty in assigning and managing doctors.

To overcome these issues, a computerized Hospital Management System is required that can automate record management and improve efficiency.

3. Project Objectives

The major objectives of this project are:

- To develop a web-based Hospital Management System.
- To store patient and doctor records in a digital database.
- To provide efficient CRUD (Create, Read, Update, Delete) operations.
- To maintain organized hospital records.
- To assign doctors to patients efficiently.
- To provide a search facility for quick record retrieval.
- To reduce paperwork and manual effort.
- To improve data accuracy and accessibility.
- To enhance hospital administrative performance.

4. Scope of the Project

The scope of the proposed system includes:

- Patient Registration and Management
- Doctor Registration and Management

- Doctor Assignment to Patients
- Patient Information Storage
- Search and Retrieval of Records
- Updating and Deleting Records
- Dashboard-Based Data Management

The system is suitable for:

- Hospitals
- Clinics
- Medical Centers
- Healthcare Organizations

5. Proposed System

The proposed Hospital Management System is a centralized web application that manages doctor and patient information through an interactive dashboard.

The system allows users to:

- Register new doctors.
- Register new patients.
- Assign doctors to patients.
- Update existing records.
- Delete unnecessary records.
- Search patients based on different criteria.
- View all hospital records from a single platform.

The system uses MongoDB as the database and Flask as the backend framework to ensure smooth communication between the user interface and the database.

6. System Features

A. Doctor Management Module

This module allows administrators to:

- Add new doctors.
- Edit doctor details.
- Delete doctor records.
- View doctor information.
- Manage doctor specializations and timings.

B. Patient Management Module

This module provides:

- Patient registration.
- Patient information management.
- Disease record maintenance.

- Contact information management.
- Patient record updates.
- Patient record deletion.

C. Doctor Assignment Module

This module helps:

- Assign doctors to patients.
- Track assigned doctors.
- Manage doctor-patient relationships.

D. Search Module

The search feature enables users to search patients using:

- Patient Name
- Disease Name
- Assigned Doctor

E. Dashboard Module

The dashboard provides:

- User-friendly interface
- Organized data presentation
- Quick navigation
- Efficient record management

7. Tools and Technologies

Frontend Technologies

- HTML5
- CSS3
- Bootstrap 5

Backend Technologies

- Python
- Flask Framework

Database

- MongoDB

Database Connectivity

- PyMongo

Development Tools

- Visual Studio Code (VS Code)

8. Functional Requirements

The system shall:

- Add doctor records.
- Update doctor records.
- Delete doctor records.
- Display doctor information.
- Register new patients.
- Update patient records.
- Delete patient records.
- Assign doctors to patients.
- Search patient records.
- Display all stored records.

9. Non-Functional Requirements

Performance

- Fast search operations.
- Quick database access.

Reliability

- Accurate storage of records.
- Consistent system performance.

Usability

- Easy-to-use interface.
- Simple navigation.

Security

- Secure database storage.
- Controlled access to records.

Scalability

- Future modules can be added easily.

10. Expected Outcomes

After successful implementation, the system will:

- Improve hospital record management.
- Reduce paperwork.
- Increase efficiency.
- Improve patient record accessibility.

- Save administrative time.
- Enhance doctor-patient management.
- Provide organized and centralized data storage.

11. Future Enhancements

The following features can be added in future versions:

- User Authentication and Login System
- Admin Panel
- Appointment Scheduling
- Billing and Payment Module
- Prescription Management
- Medical Report Generation
- Patient History Tracking
- SMS and Email Notifications
- Online Consultation System
- Cloud Database Integration

12. Conclusion

The Advanced Hospital Management System is designed to automate and simplify hospital management operations. The system provides an effective platform for managing patient and doctor records through a centralized database. By using Flask and MongoDB, the project ensures efficiency, reliability, and ease of use. The proposed solution will significantly reduce manual work, improve record accuracy, and enhance the overall management process within healthcare organizations.

Group Contribution

Member Name	Contribution
Fatima Tehreem	Backend development, Flask application development, MongoDB database integration, CRUD operations, routing, search functionality, and database management.
Waleed Ahmed	Frontend development using HTML, CSS, Bootstrap, dashboard design, responsive user interface development, forms design, and user experience improvements.
Laiba Bibi	Project analysis, testing, debugging, documentation, report writing, presentation preparation, and project coordination.

